Vicente Bastura

Mechanical Engineer

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EDUCATION

Worcester Polytechnic Institute – BS/MS in Mechanical Engineering | Undergraduate GPA – 3.82 May 2027

PROJECTS

Autonomous Off-Road Vehicle Chassis – Major Qualifying Project May 2026 - Present

- Contributed to iterative spaceframe chassis design by considering structural, packaging, and size constraints.
- Conducted static FEA simulations using Ansys to analyze the chassis for torsional rigidity, peak stresses, and load paths.
- Collaborating with other teams to create an entire off-road vehicle platform, with a full rolling chassis and support for future systems.

1 Pound Plastic Combat Robot October 2025 - Present

- Designed a parametric weapon, iterated mechanical components, and collaboratively brainstormed for version 1.
- Manufactured the complete robot using 3D printing and wired the electronic system for the first competition.
- Identified weaknesses in workflow, material, and design considerations for version 2 development.
- Redesigned the failed weapon to be asymmetric and smaller for version 2.

MIDI Music Box August 2025 - October 2025

- Collaborated on the mechanical design, assembly of custom 3D-printed actuators, and an electronic system connected to a modified music box.
- Wired and integrated 20 servos with servo driver boards to independently actuate individual music box tines.
- Developed Arduino code to parse MIDI input and map incoming notes to the corresponding servos.
- Implemented individual servo position offsets in software to compensate for variation in the assembly and ensure reliable note actuation.

Understanding Perceptions of Cuenca Cantón Wildfires – Interactive Qualifying Project October 2025 - March 2026

- Collaborated with a 4-person team and the University of Cuenca to investigate community perceptions of wildfire impacts on soil in Cuenca, Ecuador.
- Interviewed rural and municipal stakeholders alongside University of Cuenca students, administering surveys across two distinct populations.
- Performed statistical analysis and authored the results section, applying chi-squared testing with Holm-Bonferroni correction across 61 respondents' survey data.

SKILLS

CAD and Design: SolidWorks (CSWA), SolidWorks Simulation, Ansys, Fusion 360, Onshape, 3D Part & Assembly Design, 2D Drawings
Manufacturing and Prototyping: 3D Printing (Operation & Maintenance), Rapid Prototyping, CAM, CNC Mill, CNC Lathe, Calipers, Micrometer, Measurement Devices, Understanding of Manufacturing Processes, OSHA-10 Certified
Programming and Electronics: MATLAB, Arduino (C++), Raspberry Pi, Sensors, Actuators, Soldering
Engineering Practice & Other: Teamwork, Hands-On, Problem Solving, Attention to Detail, Creative, Communication

EXPERIENCE

Landscaping, Multiple Companies – Pomfret, CT May 2024 – August 2026

- Managed a private client's property independently, scheduling work, and determining maintenance needs while completing additional assigned projects including pathway and retaining-wall construction. (2024-2026)
- Completed daily tasks independently at a golf course, operating push mowers, weed trimmers, leaf blowers, and specialty equipment. (2024)
- Maintained residential properties for a landscaping company, traveling between job sites and weed trimming. (2026)